

LAPORAN TUGAS BESAR PRAKTIKUM DASAR PEMROGRAMAN



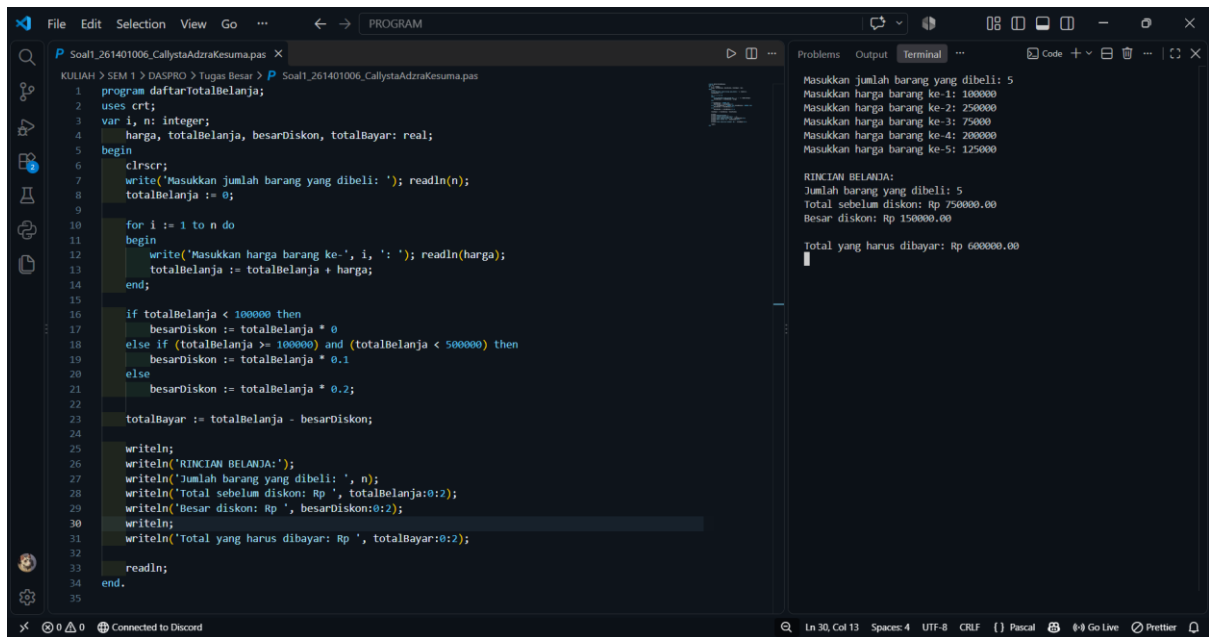
Disusun oleh:

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Laboratorium Dasar Pemrograman 5

Soal 1



The screenshot shows a Pascal program in a code editor. The program is titled 'Soal1_261401006_CallystaAdzraKesuma.pas'. It implements a shopping cart system with the following logic: it prompts the user for the number of items (n), then for each item's price (harga). It calculates the total price (totalBelanja) and applies a discount (besarDiskon) based on the total price: 0% for total < 100,000, 10% for 100,000 <= total < 500,000, and 20% for total >= 500,000. The final amount to be paid (totalBayar) is the total price minus the discount. The program then displays a summary of the purchase.

```
1 program daftarTotalBelanja;
2 uses crt;
3 var i, n: integer;
4     harga, totalBelanja, besarDiskon, totalBayar: real;
5 begin
6     clrscr;
7     write('Masukkan jumlah barang yang dibeli: '); readln(n);
8     totalBelanja := 0;
9
10    for i := 1 to n do
11    begin
12        write('Masukkan harga barang ke-', i, ': '); readln(harga);
13        totalBelanja := totalBelanja + harga;
14    end;
15
16    if totalBelanja < 100000 then
17        besarDiskon := totalBelanja * 0
18    else if (totalBelanja >= 100000) and (totalBelanja < 500000) then
19        besarDiskon := totalBelanja * 0.1
20    else
21        besarDiskon := totalBelanja * 0.2;
22    totalBayar := totalBelanja - besarDiskon;
23
24    writeln;
25    writeln('RINCIAN BELANJA:');
26    writeln('Jumlah barang yang dibeli: ', n);
27    writeln('Total sebelum diskon: Rp ', totalBelanja:0:2);
28    writeln('Besar diskon: Rp ', besarDiskon:0:2);
29    writeln;
30    writeln('Total yang harus dibayar: Rp ', totalBayar:0:2);
31
32    readln;
33 end.
```

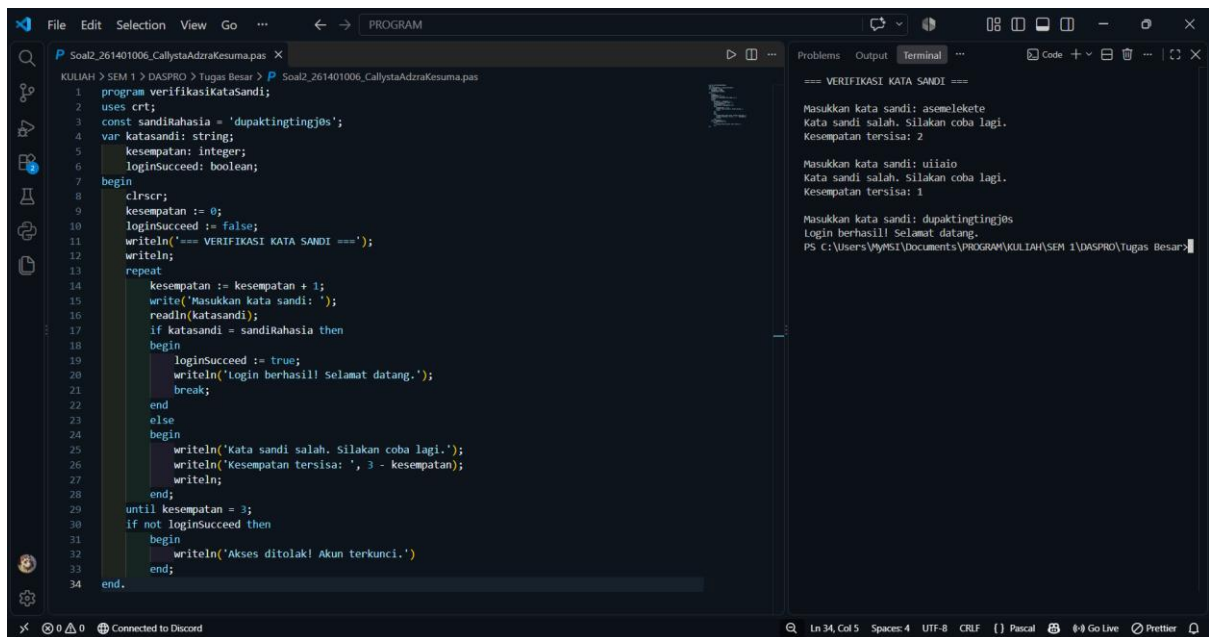
The terminal output shows the program's execution with the following input and output:

```
Masukkan jumlah barang yang dibeli: 5
Masukkan harga barang ke-1: 100000
Masukkan harga barang ke-2: 250000
Masukkan harga barang ke-3: 75000
Masukkan harga barang ke-4: 200000
Masukkan harga barang ke-5: 125000

RINCIAN BELANJA:
Jumlah barang yang dibeli: 5
Total sebelum diskon: Rp 750000.00
Besar diskon: Rp 150000.00

Total yang harus dibayar: Rp 600000.00
```

Soal 2



The screenshot shows a Pascal program in a code editor. The program is titled 'Soal2_261401006_CallystaAdzraKesuma.pas'. It implements a password verification system with the following logic: it prompts the user to enter a password (kataSandi) and compares it with a predefined password (sandirahasia). If the passwords match, the user is logged in successfully. If not, the user is prompted to try again, up to 3 attempts. If the user fails to log in after 3 attempts, the program displays a message: 'Akses ditolak! Akun terkunci.'.

```
1 program verifikasiKataSandi;
2 uses crt;
3 const sandirahasia = 'dupaktingtingj0s';
4 var katasandi: string;
5     kesempatan: integer;
6     loginSucceed: boolean;
7 begin
8     clrscr;
9     kesempatan := 0;
10    loginSucceed := false;
11    writeln('=== VERIFIKASI KATA SANDI ===');
12    writeln;
13    repeat
14        kesempatan := kesempatan + 1;
15        write('Masukkan kata sandi: ');
16        readln(katasandi);
17        if katasandi = sandirahasia then
18        begin
19            loginSucceed := true;
20            writeln('Login berhasil! Selamat datang.');

The terminal output shows the program's execution with the following input and output:



```
=== VERIFIKASI KATA SANDI ===

Masukkan kata sandi: asemelelete
Kata sandi salah. Silakan coba lagi.
Kesempatan tersisa: 2

Masukkan kata sandi: uiiaio
Kata sandi salah. Silakan coba lagi.
Kesempatan tersisa: 1

Masukkan kata sandi: dupaktingtingj0s
Login berhasil! Selamat datang.
PS C:\Users\MyMSI\Documents\PROGRAM\KULIAH\SEM 1\DASPRO\Tugas Besar>
```


```

Soal 3

```
1 program DeretAngkaFilter;
2 uses crt;
3 var
4     n, pilihan, i: integer;
5 begin
6     clrscr;
7     write('Masukkan nilai N: ');
8     readln(n);
9     writeln('Pilih kategori deret:');
10    writeln('1. Ganjil');
11    writeln('2. Genap');
12    write('Masukkan pilihan (1/2): ');
13    readln(pilihan);
14    writeln;
15    writeln('Hasil deret angka:');
16
17    i := 1;
18    while i <= n do
19    begin
20        if ((pilihan = 1) and (i mod 2 = 0)) or
21            ((pilihan = 2) and (i mod 2 <> 0)) or
22            (i mod 5 = 0) then
23        begin
24            i := i + 1;
25            continue;
26        end;
27        write(i, ' ');
28        i := i + 1;
29    end;
30    writeln;
31    readln;
32 end.
```

Masukkan nilai N: 10
Pilih kategori deret:
1. Ganjil
2. Genap
Masukkan pilihan (1/2): 2

Hasil deret angka:
2 4 6 8

Masukkan nilai N: 10
Pilih kategori deret:
1. Ganjil
2. Genap
Masukkan pilihan (1/2): 1

Hasil deret angka:
1 3 7 9

Soal 4

```
1 program Kalkulator;  
2 uses crt;  
3 var  
4     pilihan: integer;  
5     angka1, angka2, hasil: real;  
6     bilangan1, bilangan2: integer;  
7     ulang: char;  
8 begin  
9     clrscr;  
10    repeat  
11        writeln('=== KALKULATOR SEDERHANA ===');  
12        writeln('1. Penjumlahan');  
13        writeln('2. Pengurangan');  
14        writeln('3. Perkalian');  
15        writeln('4. Pembagian Real');  
16        writeln('5. DIV & MOD');  
17        writeln('=====');  
18        write('Pilih operasi (1-5): ');  
19        readln(pilihan);  
20        case pilihan of  
21            1:  
22                begin  
23                    write('Masukkan angka pertama: ');  
24                    readln(angka1);  
25                    write('Masukkan angka kedua: ');  
26                    readln(angka2);  
27                    hasil := angka1 + angka2;  
28                    writeln('Hasil = ', hasil:0:2);  
29                end;  
30            2:  
31                begin  
32                    write('Masukkan angka pertama: ');  
33                    readln(angka1);  
34                    write('Masukkan angka kedua: ');  
35                    readln(angka2);  
36                    hasil := angka1 - angka2;  
37                    writeln('Hasil = ', hasil:0:2);  
38                end;  
39            3:  
40                begin  
41                    write('Masukkan angka pertama: ');  
42                    readln(angka1);  
43                    write('Masukkan angka kedua: ');  
44                    readln(angka2);  
45                    hasil := angka1 * angka2;  
46                    writeln('Hasil = ', hasil:0:2);  
47                end;  
48            4:  
49                begin  
50                    write('Masukkan angka pertama: ');  
51                    readln(angka1);  
52                    write('Masukkan angka kedua: ');  
53                    readln(angka2);  
54                    if angka2 = 0 then  
55                        writeln('Tidak dapat melakukan pembagian dengan 0.')56                    else  
57                        begin  
58                            hasil := angka1 / angka2;  
59                            writeln('Hasil = ', hasil:0:2);  
60                        end;  
61                    end;  
62                end;  
63            5:  
64                begin  
65                    write('Masukkan bilangan pertama: ');  
66                    readln(bilangan1);  
67                    write('Masukkan bilangan kedua: ');  
68                    readln(bilangan2);  
69                    if bilangan2 = 0 then  
70                        writeln('Tidak dapat melakukan DIV dan MOD dengan 0.')71                    else  
72                        begin  
73                            writeln('Hasil DIV = ', bilangan1 div bilangan2);  
74                            writeln('Hasil MOD = ', bilangan1 mod bilangan2);  
75                        end;  
76                    end;  
77                end;  
78        end;  
79        write('Apakah ingin melakukan perhitungan lagi? (Y/T): ');  
80        readln(ulang);  
81        until (ulang = 'T') or (ulang = 't');  
82        writeln('Program selesai.');
```

Soal 5

```
File Edit Selection View Go ... PROGRAM
P Soal5_261401006_CallystaAdzraKesuma.pas
KULIAH > SEM 1 > DASPRO > Tugas Besar > P Soal5_261401006_CallystaAdzraKesuma.pas
1 program RekapitulasiNilai;
2 uses crt;
3 var
4   m, n, i, j, lulus, tidaklulus: integer;
5   nilai, totalNilai, rataRata: real;
6 begin
7   clrscr;
8   write('Masukkan jumlah mahasiswa: ');
9   readln(m);
10  write('Masukkan jumlah tugas: ');
11  readln(n);
12  lulus := 0; tidaklulus := 0;
13  for i := 1 to m do
14    begin
15      writeln;
16      write('--- Input Data Mahasiswa ke-', i, ' ---');
17      totalNilai := 0;
18      for j := 1 to n do
19        begin
20          write('Masukkan nilai tugas ke-', j, ': ');
21          readln(nilai);
22          totalNilai := totalNilai + nilai;
23        end;
24      rataRata := totalNilai / n;
25      write('Rata-rata Nilai: ', rataRata:0:2);
26      if rataRata >= 65 then
27        begin
28          writeln(' [LULUS]');
29          lulus := lulus + 1;
30        end
31      else
32        begin
33          writeln(' [TIDAK LULUS]');
34          tidaklulus := tidaklulus + 1;
35        end;
36      end;
37      writeln;
38      writeln('RINGKASAN HASIL REKAPITULASI NILAI MAHASISWA');
39      writeln('Total mahasiswa LULUS : ', lulus);
40      writeln('Total mahasiswa TIDAK LULUS : ', tidaklulus);
41      readln;
42    end;
43 end.
```

Problems Output Terminal ... Code + - - - - -

```
Masukkan jumlah mahasiswa: 5
Masukkan jumlah tugas: 3

--- Input Data Mahasiswa ke-1 ---
Masukkan nilai tugas ke-1: 80
Masukkan nilai tugas ke-2: 75
Masukkan nilai tugas ke-3: 50
Rata-rata Nilai: 81.67 [LULUS]

--- Input Data Mahasiswa ke-2 ---
Masukkan nilai tugas ke-1: 0
Masukkan nilai tugas ke-2: 75
Masukkan nilai tugas ke-3: 98
Rata-rata Nilai: 57.67 [TIDAK LULUS]

--- Input Data Mahasiswa ke-3 ---
Masukkan nilai tugas ke-1: 80
Masukkan nilai tugas ke-2: 97
Masukkan nilai tugas ke-3: 89
Rata-rata Nilai: 88.67 [LULUS]

--- Input Data Mahasiswa ke-4 ---
Masukkan nilai tugas ke-1: 96
Masukkan nilai tugas ke-2: 90
Masukkan nilai tugas ke-3: 89
Rata-rata Nilai: 91.67 [LULUS]

--- Input Data Mahasiswa ke-5 ---
Masukkan nilai tugas ke-1: 55
Masukkan nilai tugas ke-2: 65
Masukkan nilai tugas ke-3: 87
Rata-rata Nilai: 69.00 [LULUS]

RINGKASAN HASIL REKAPITULASI NILAI MAHASISWA
Total mahasiswa LULUS : 4
Total mahasiswa TIDAK LULUS : 1
```

Soal 6

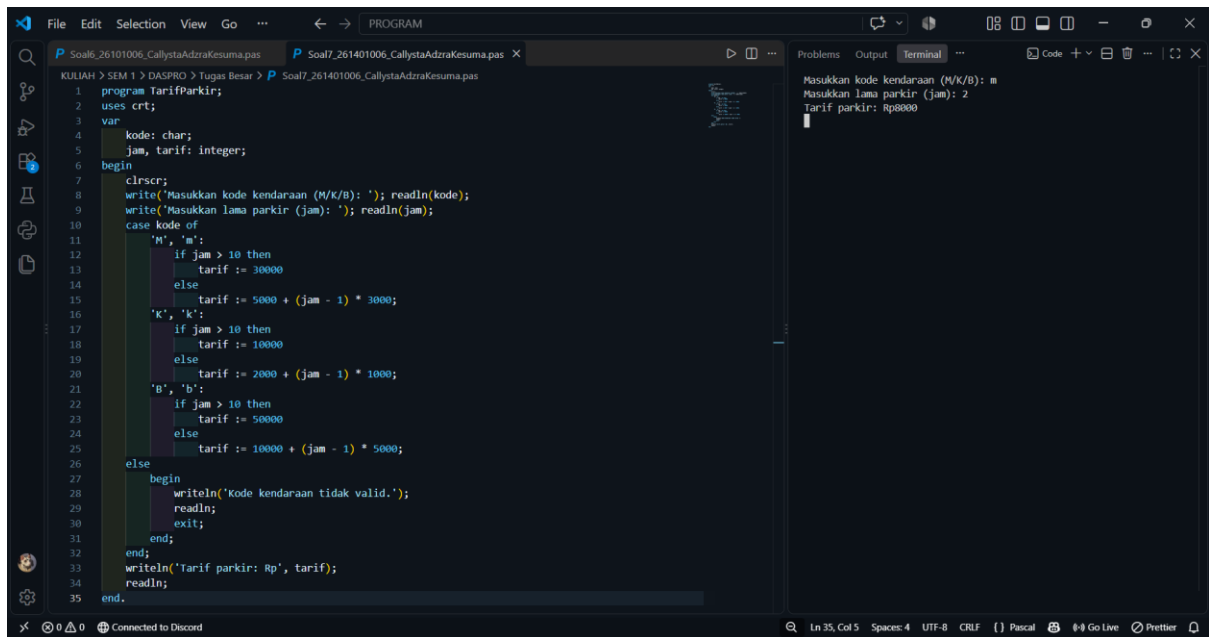
```
File Edit Selection View Go ... PROGRAM
P Soal6_26101006_CallystaAdzraKesuma.pas
KULIAH > SEM 1 > DASPRO > Tugas Besar > P Soal6_26101006_CallystaAdzraKesuma.pas
1 program PenentuanNilaiAkhir;
2 uses crt;
3 var
4   tugas, uts, uas, nilaiAkhir: real;
5   kehadiran: integer;
6   indeks: char;
7   status: string;
8 begin
9   clrscr;
10  write('Masukkan nilai Tugas : '); readln(tugas);
11  write('Masukkan nilai UTS : '); readln(uts);
12  write('Masukkan nilai UAS : '); readln(uas);
13  write('Masukkan % Kehadiran : '); readln(kehadiran);
14  nilaiAkhir := (0.3 * tugas) + (0.3 * uts) + (0.4 * uas);
15  if (nilaiAkhir >= 60) and (kehadiran >= 80) then
16    status := 'LULUS'
17  else
18    status := 'TIDAK LULUS';
19  if nilaiAkhir >= 85 then indeks := 'A'
20  else if nilaiAkhir >= 75 then indeks := 'B'
21  else if nilaiAkhir >= 60 then indeks := 'C'
22  else if nilaiAkhir >= 50 then indeks := 'D'
23  else indeks := 'E';
24  writeln;
25  writeln('Hasil Perhitungan Nilai Akhir');
26  writeln('Nilai Akhir : ', nilaiAkhir:0:2);
27  writeln('Indeks : ', indeks);
28  writeln('Status : ', status);
29  readln;
30 end;
31
```

Problems Output Terminal ... Code + - - - - -

```
Masukkan nilai Tugas : 85
Masukkan nilai UTS : 90
Masukkan nilai UAS : 89
Masukkan % Kehadiran : 95

Hasil Perhitungan Nilai Akhir
Nilai Akhir : 88.10
Indeks : A
Status : LULUS
```

Soal 7



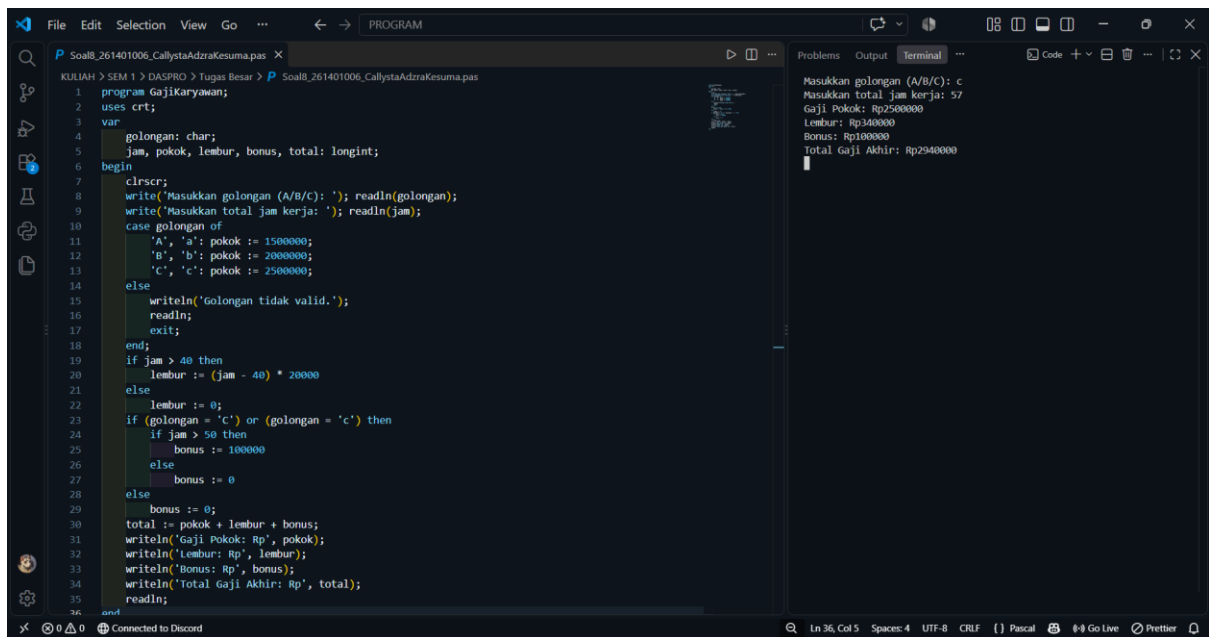
The screenshot shows a Pascal IDE with a program for calculating parking fees. The program prompts the user for a vehicle type (M/K/B) and parking duration (jam). It uses a case statement to determine the fee based on the vehicle type and duration. The output shows the calculated fee for a motor vehicle (M) parked for 2 hours, resulting in Rp0000.

```
1 program TariffParkir;  
2 uses crt;  
3 var  
4     kode: char;  
5     jam, tarif: integer;  
6 begin  
7     clrscr;  
8     write('Masukkan kode kendaraan (M/K/B): '); readln(kode);  
9     write('Masukkan lama parkir (jam): '); readln(jam);  
10    case kode of  
11        'M', 'm':  
12            if jam > 10 then  
13                tarif := 30000  
14            else  
15                tarif := 5000 + (jam - 1) * 3000;  
16        'K', 'k':  
17            if jam > 10 then  
18                tarif := 10000  
19            else  
20                tarif := 2000 + (jam - 1) * 1000;  
21        'B', 'b':  
22            if jam > 10 then  
23                tarif := 50000  
24            else  
25                tarif := 10000 + (jam - 1) * 5000;  
26    else  
27        begin  
28            writeln('Kode kendaraan tidak valid.');29            readln;  
30            exit;  
31        end;  
32    end;  
33    writeln('Tarif parkir: Rp', tarif);  
34    readln;  
35 end.
```

Output:

```
Masukkan kode kendaraan (M/K/B): m  
Masukkan lama parkir (jam): 2  
Tarif parkir: Rp0000
```

Soal 8



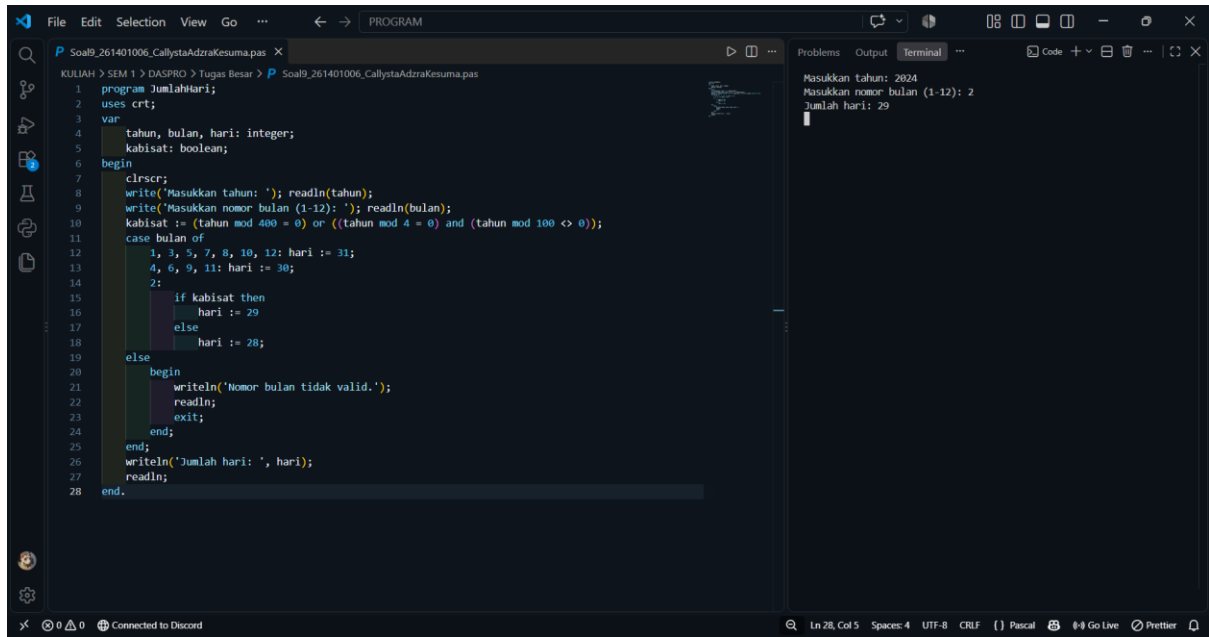
The screenshot shows a Pascal IDE with a program for calculating salary. The program prompts the user for a group (A/B/C) and total working hours (jam). It uses a case statement to determine the base salary (pokok) based on the group. It then calculates overtime (lembur) and bonus based on the total working hours. The output shows the calculated salary components for group A with 57 hours of work, resulting in a total salary of Rp2940000.

```
1 program GajiKaryawan;  
2 uses crt;  
3 var  
4     golongan: char;  
5     jam, pokok, lembur, bonus, total: longint;  
6 begin  
7     clrscr;  
8     write('Masukkan golongan (A/B/C): '); readln(golongan);  
9     write('Masukkan total jam kerja: '); readln(jam);  
10    case golongan of  
11        'A', 'a': pokok := 1500000;  
12        'B', 'b': pokok := 2000000;  
13        'C', 'c': pokok := 2500000;  
14    else  
15        writeln('Golongan tidak valid.');16        readln;  
17        exit;  
18    end;  
19    if jam > 40 then  
20        lembur := (jam - 40) * 20000  
21    else  
22        lembur := 0;  
23    if (golongan = 'C') or (golongan = 'c') then  
24        if jam > 50 then  
25            bonus := 100000  
26        else  
27            bonus := 0  
28        else  
29            bonus := 0;  
30    total := pokok + lembur + bonus;  
31    writeln('Gaji Pokok: Rp', pokok);  
32    writeln('Lembur: Rp', lembur);  
33    writeln('Bonus: Rp', bonus);  
34    writeln('Total Gaji Akhir: Rp', total);  
35    readln;  
36 end.
```

Output:

```
Masukkan golongan (A/B/C): c  
Masukkan total jam kerja: 57  
Gaji Pokok: Rp2500000  
Lembur: Rp340000  
Bonus: Rp100000  
Total Gaji Akhir: Rp2940000
```

Soal 9

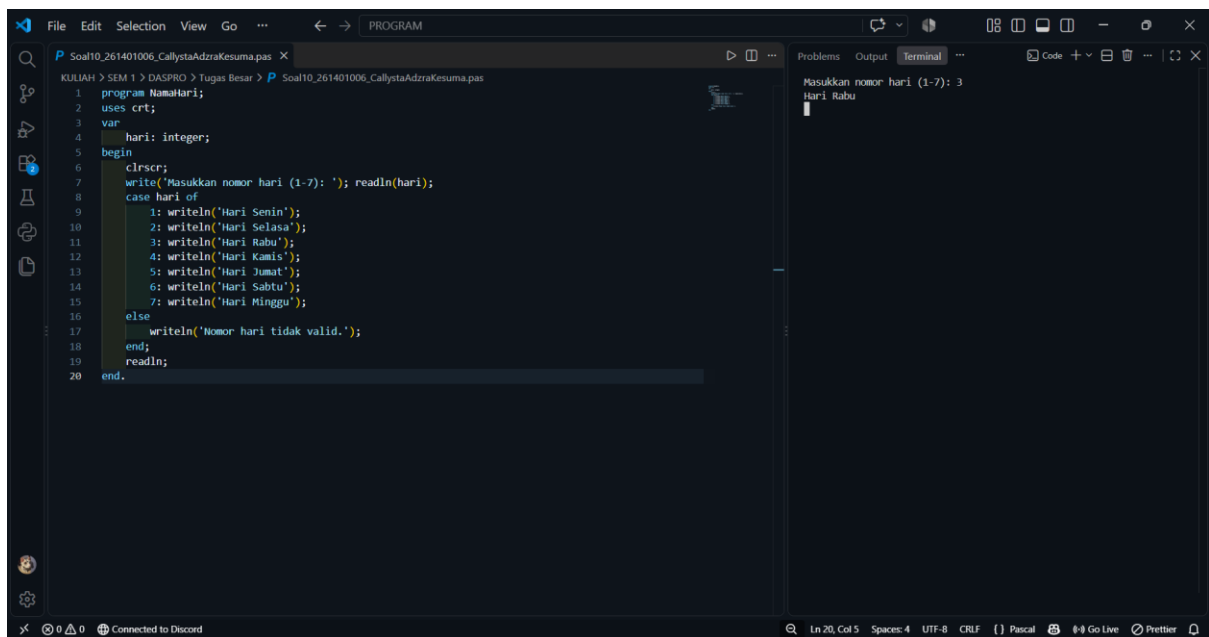


```
1 program JumlahHari;
2 uses crt;
3 var
4     tahun, bulan, hari: integer;
5     kabisat: boolean;
6 begin
7     clrscr;
8     write('Masukkan tahun: '); readln(tahun);
9     write('Masukkan nomor bulan (1-12): '); readln(bulan);
10    kabisat := (tahun mod 400 = 0) or ((tahun mod 4 = 0) and (tahun mod 100 <> 0));
11    case bulan of
12        1, 3, 5, 7, 8, 10, 12: hari := 31;
13        4, 6, 9, 11: hari := 30;
14        2:
15            if kabisat then
16                hari := 29
17            else
18                hari := 28;
19    else
20        begin
21            writeln('Nomor bulan tidak valid.');

Masukkan tahun: 2024  
Masukkan nomor bulan (1-12): 2  
Jumlah hari: 29


```

Soal 10



```
1 program NamaHari;
2 uses crt;
3 var
4     hari: integer;
5 begin
6     clrscr;
7     write('Masukkan nomor hari (1-7): '); readln(hari);
8     case hari of
9         1: writeln('Hari Senin');
10        2: writeln('Hari Selasa');
11        3: writeln('Hari Rabu');
12        4: writeln('Hari Kamis');
13        5: writeln('Hari Jumat');
14        6: writeln('Hari Sabtu');
15        7: writeln('Hari Minggu');
16    else
17        writeln('Nomor hari tidak valid.');

Masukkan nomor hari (1-7): 3  
Hari Rabu


```